

15. FACE : DEVELOPMENT OF THE TONGUE AND THYROID

DURATION : 07:43

STUDY SHEET

COURSE SUMMARY

This video explores the complex developmental processes of the tongue and thyroid during embryogenesis, focusing on embryonic straightening and its implications for postural dynamics. Practices such as tongue stabilisation and interaction with the thyroid cartilage are discussed, highlighting the importance of the structures surrounding the foramen caecum. The instructors also examine the impact of the branchial arches and embryonic tissues on overall development, as well as the importance of fulcrums in the body, particularly through the pelvis. These concepts are crucial for practitioners seeking to integrate a holistic approach into their osteopathic work.

DEVELOPMENT OF THE TONGUE AND THYROID

At the time of the embryo's **flexion** and **extension**, the digestive tract is initially positioned quite low. Observing the **palate**, we notice that it has an ogival shape, whereas previously it was rather low. This extension is accompanied by a **descent of the pterygoids** and a traction that causes the digestive tract to rise.

This developmental process is crucial for the formation of the **tongue** and the **thyroid**, which originates from the tongue. In the embryonic section, we observe the **endodermal** part as well as the numbered pharyngeal arches (1, 2, 3, 4). In pharyngeal arch number 1, at the time of extension, the **lateral tongue layer** appears. Between branchial arch number 2 and this layer, an **unpaired tuberculum** and a small hole, the **foramen caecum**, are formed.

The foramen caecum, which develops with the tongue, is essential for the formation of the thyroid system. This development is marked by **overall growth**, with a fixed point acting as a fulcrum. The depth of the tongue is linked to the **depth of encephalisation**.

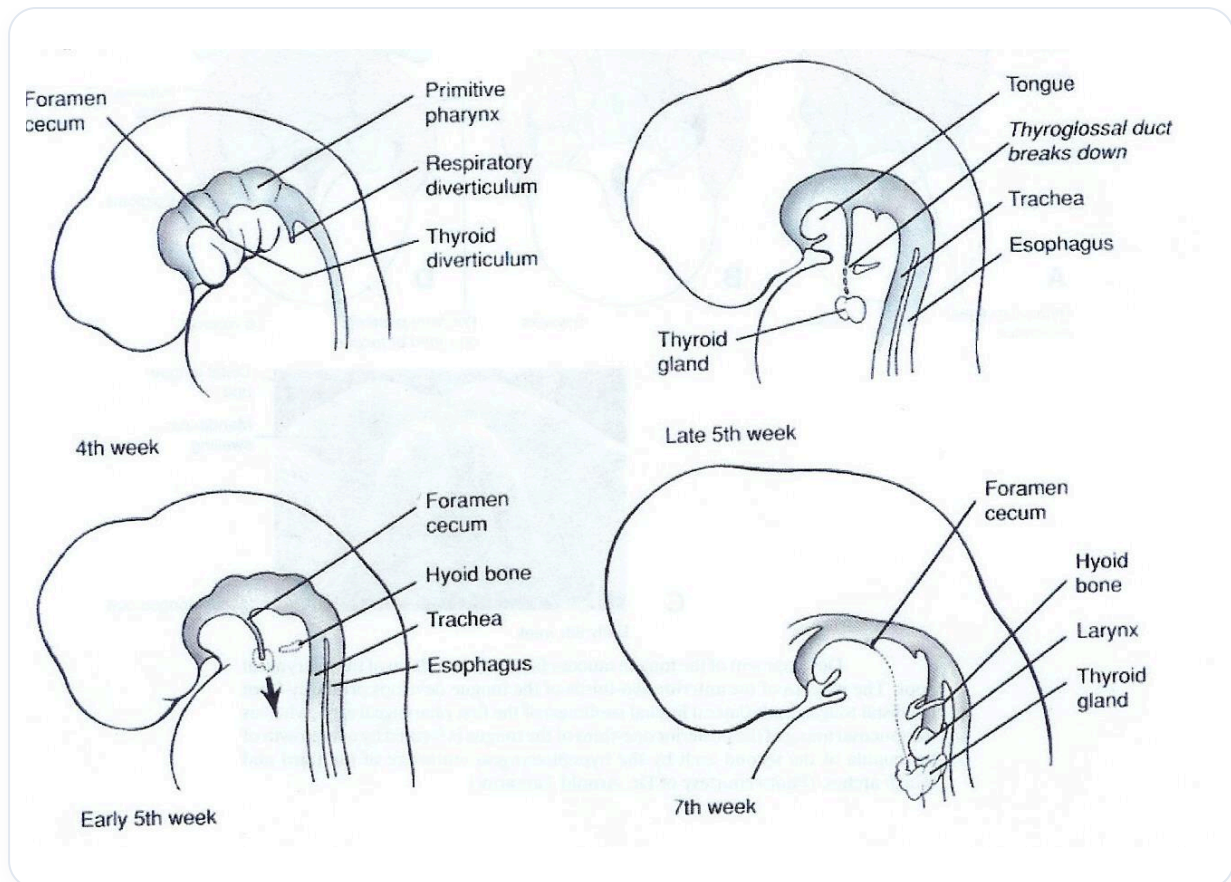


FIG. — Embryonic thyroid development

Embryonic extension reveals the foramen caecum, which is crucial for the formation of surrounding structures. Another section shows **Meckel's cartilage** and the tongue. It is important to note that the arrows in the illustrations must be oriented correctly to understand the dynamics of this development.

There is a tearing between the tongue and the diaphragm, where the **thyroid cartilage** becomes a fulcrum. Stabilising the tongue and the vague air is essential to maintain balance at the level of the thyroid cartilage and the hyoid bone. The tongue plays a fundamental role in postural activity, and the child begins to straighten up with the appearance of their first teeth, using their tongue as a fulcrum.

The tongue seeks to **plug small holes** during swallowing, and its movement can influence the longitudinal axis of the body, extending to the **pelvis**. Pelvic deviations can be a reflection of occlusal imbalance, linked to lingual dynamics.

Teeth are areas of **densification** and can retain memories, which influences body dynamics. In this phase of development, we also observe the **palatine tonsils**, the **parathyroid**, the **thymus**, and the **ultimobranchial body**. Each branchial arch (mandibular, thyroid, superior and inferior laryngeal) is interconnected, and the endodermal tissue plays a central role in this dynamic.

Embryonic development takes place over a specific period, with the **closure of the neural tube** on the 27th day and the complete extension of the embryo after the first month. The formation of the embryo is completed between the second and third month, with well-defined structures.

Embryonic tissues influence each other: the **ectoderm** and **endoderm** depend on the **mesoderm**, which ensures the transmission of nutritional information. This dynamic is essential for understanding the development of the tongue, thyroid, and parathyroid.

It is crucial to follow the **fulcrums** in the body, as they manifest in practice. The **navel** is a major fulcrum, and it is essential that the pelvis is clear and free for proper functioning. In osteopathic practice, treating the pelvis is often a priority, as it is linked to the **sphenobasilar synchondrosis**. Working on the sacrum or pelvis directly influences this synchondrosis, thus allowing a holistic approach to the body.

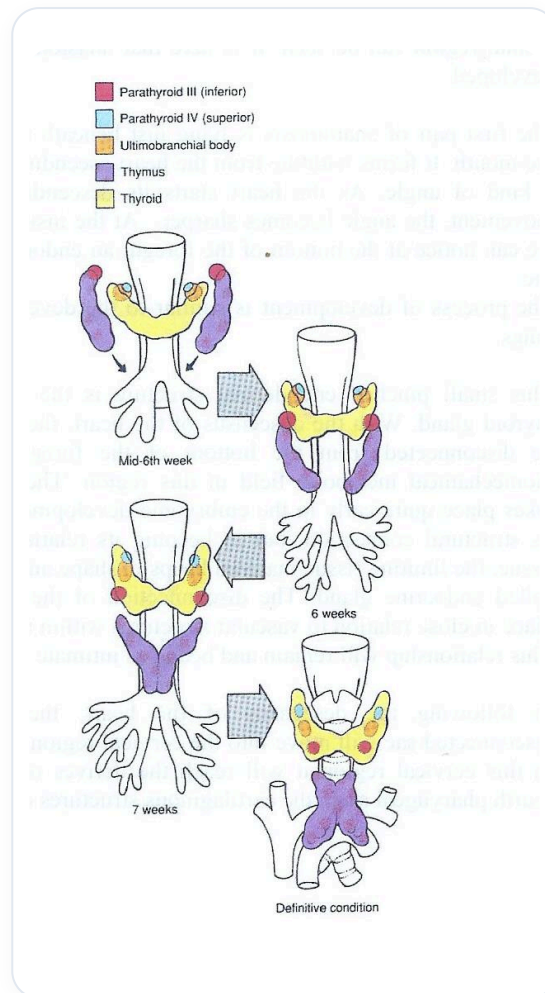


FIG. — *Thyroid and parathyroid development*

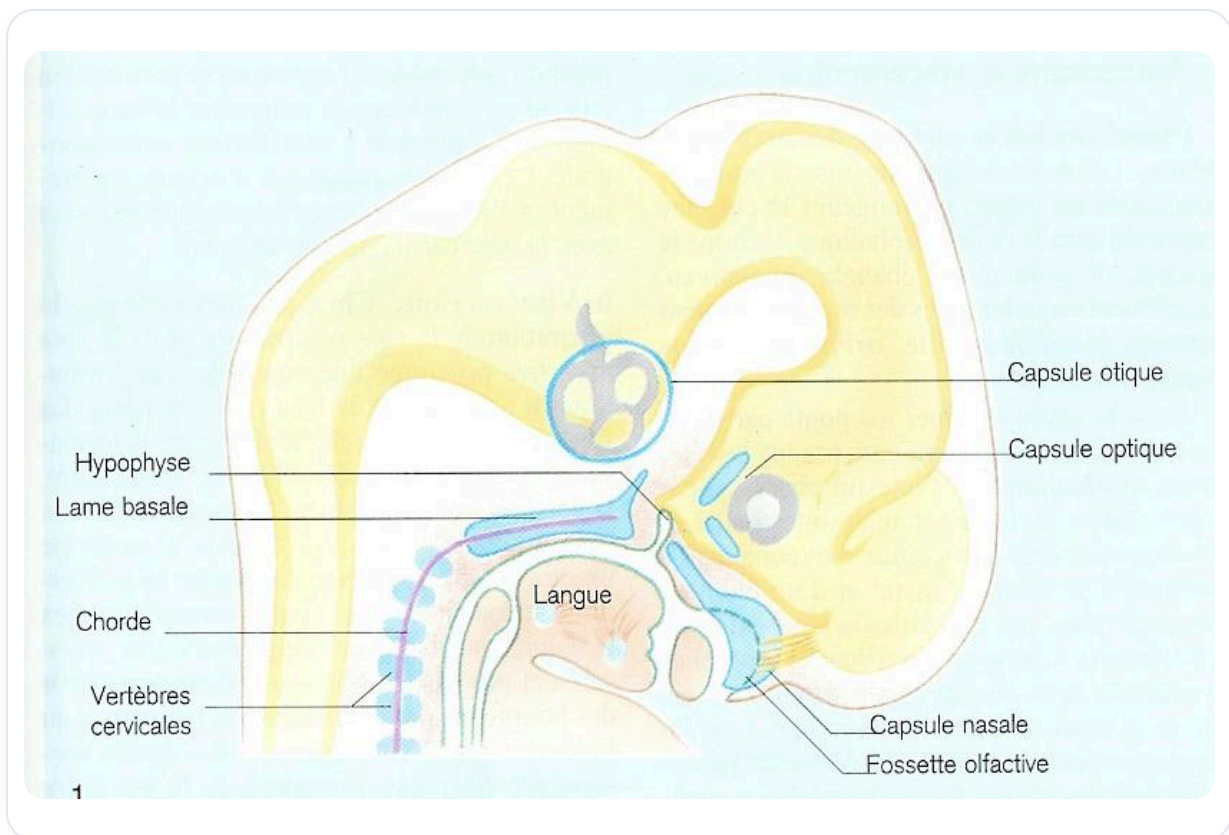


FIG. — *Human embryo sagittal*

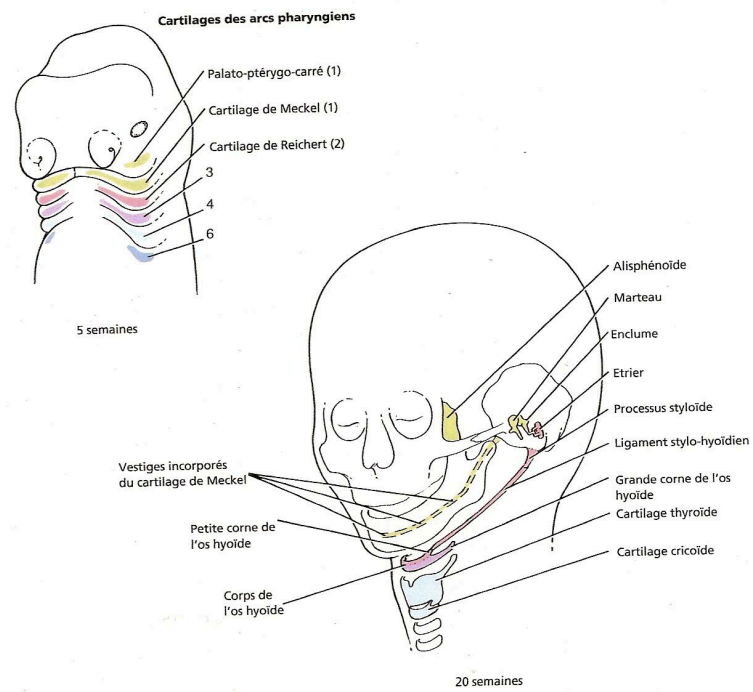


Fig. 12 - 5. Devenir des cartilages des arcs pharyngiens. Ces cartilages donnent un petit os de l'orbite, les éléments squelettiques des mâchoires, les trois osselets de l'oreille, l'os hyoïde et le squelette du larynx.

FIG. — Pharyngeal arch development

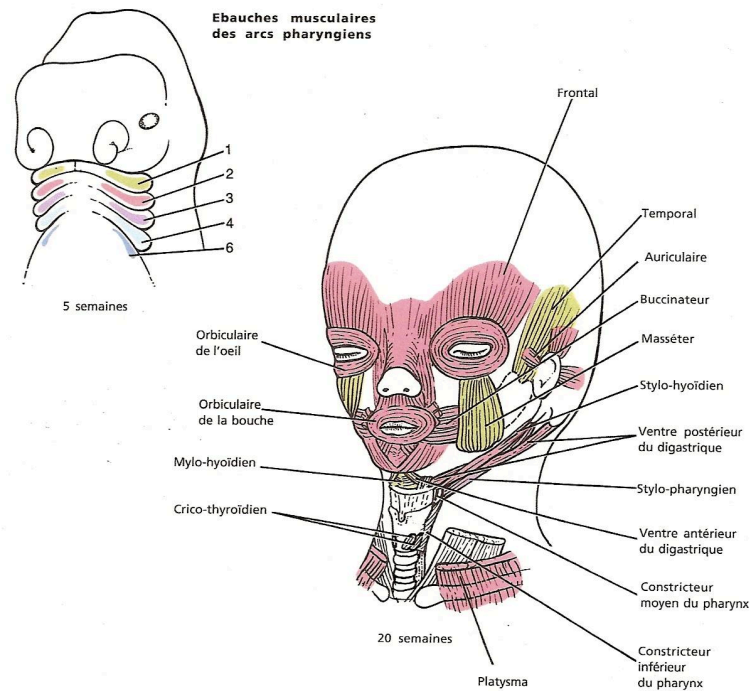


Fig. 12 - 7. Devenir de la musculature des arcs pharyngiens. Les muscles des arcs pharyngiens se développent à partir du mésoblaste para-axial qui se constitue principalement à partir des somitomères crâniens et des somites occipitaux. Les myoblastes du sixième arc fournissent la musculature intrinsèque du larynx (non représentée).

FIG. — Pharyngeal arch musculature

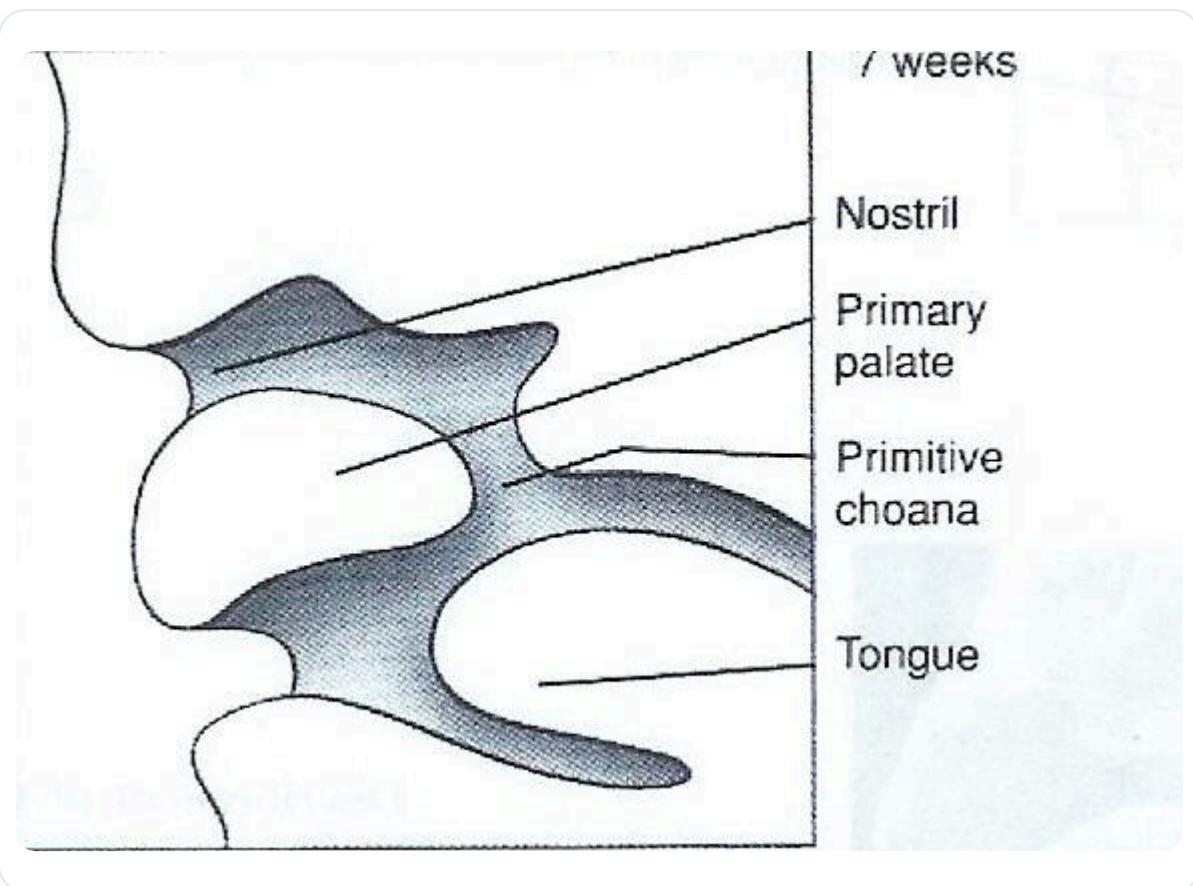


FIG. — Palate development embryo

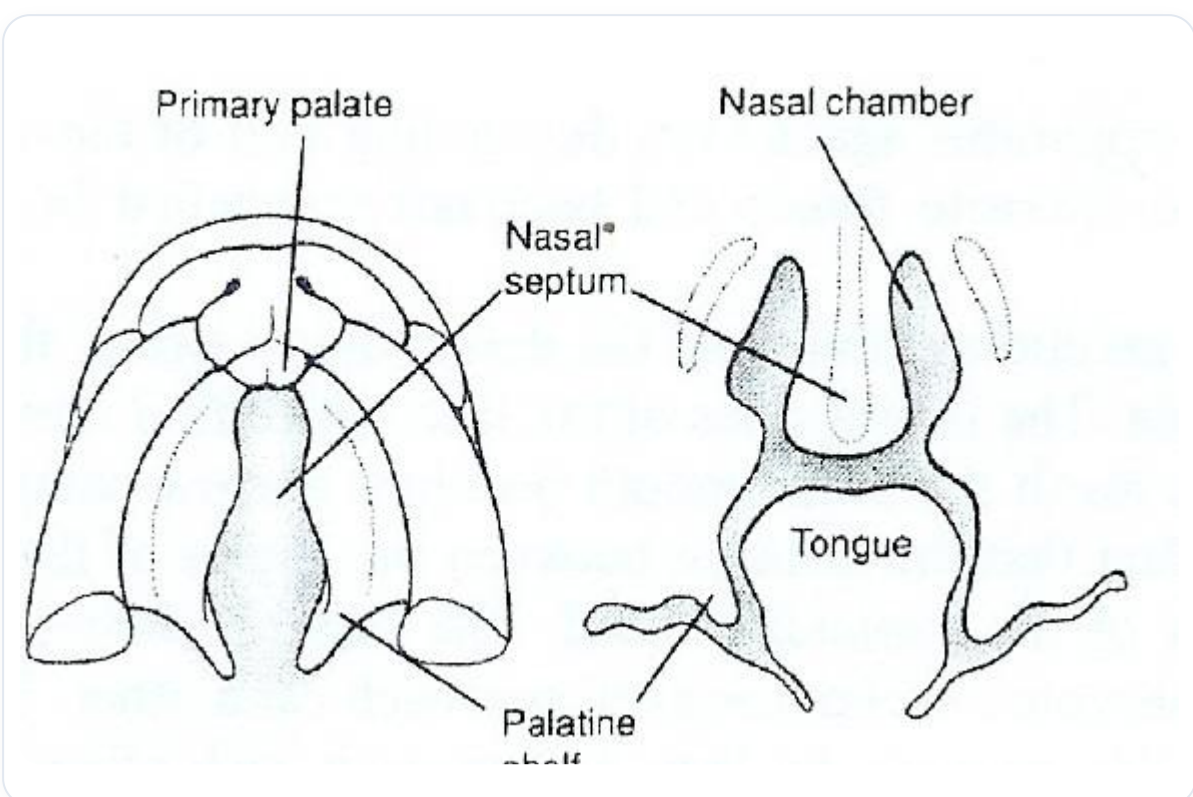


FIG. — Palate and nose development

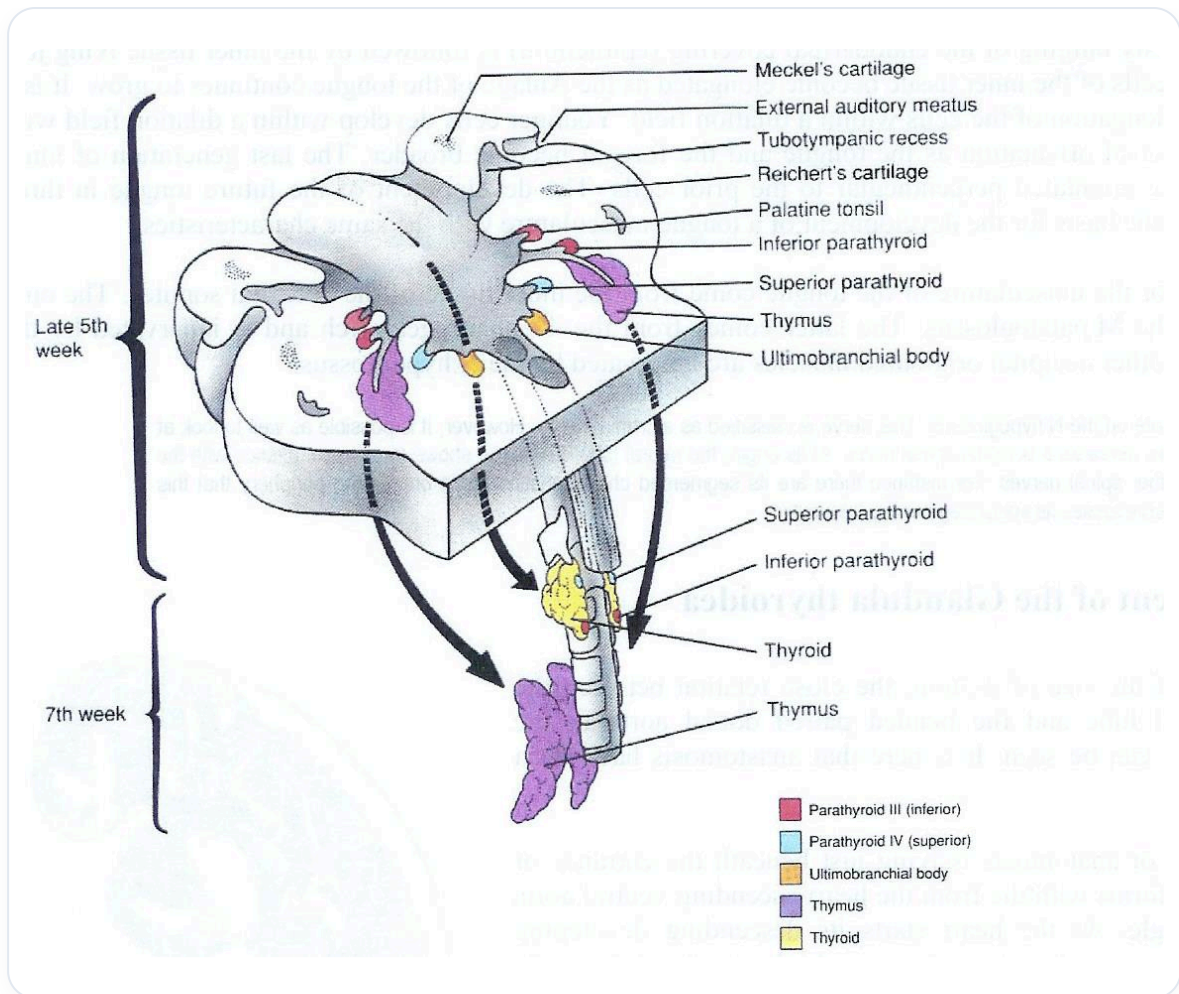


FIG. — Pharyngeal development